Conclusion

EduGenie - AI Learning Assistant

1. Summary of the Project

EduGenie is a multi-model AI learning portal built with FastAPI, Jinja2, HTML and CSS. A student opens one page, chooses a task, enters a question, topic or text, and receives a result together with the name of the model that produced it. The portal offers five study aids: concept explanation, question answering, summarisation, quiz generation and learning-path creation.

The project is built on a hybrid model strategy. The simplest task, concept explanation, runs on the local LaMini-Flan-T5-783M model, so it has no per-request cost. The four tasks that need longer or more strictly structured output run on Gemini through the google-genai client.

2. How the System Was Built

Stage	What was done	Benefit
Model Selection and Architecture	Chose the local + cloud model mix and a layered, one-module-per-task structure	Low cost, clear code, easy to extend
Core Functionalities Development	Built five task modules with prompts, validation, a retry helper, routes and error handling	Each learning tool works independently and fails safely
Frontend Development	Built index.html and style.css with a task selector, query box and result box with model badge	Simple page that anyone can use
Testing and Deployment	Defined test cases, route tests and local/production deployment steps	Repeatable checking and set-up

3. Objectives Achieved

Objective	Outcome
Explain concepts quickly	Local model gives a short explanation with no API cost.
Answer questions in depth	qna.py requests detailed answers with practical examples.
Save reading time	summary_module.py produces an overview and bullet takeaways.
Test understanding	quiz_module.py creates a 3-question quiz with an Answer Key.
Plan long-term study	learning_path.py produces a phased roadmap with projects and timelines.
Stay reliable	Start-up key checks, empty-input checks, retry helper and a route-level error handler keep the page working.
Be transparent	The drop-down and the answer badge always name the model.

4. Key Strengths

- **Modular design:** adding or changing a feature touches one module.
- **Hybrid models:** cost, speed and quality are balanced by using each model where it fits best.

- **Safe configuration:** the key comes from the environment and is validated at start-up.
- **Efficient resource use:** the local model is loaded only when first needed and then reused.
- **Readable results:** the response box keeps the line breaks of quizzes and roadmaps.
- **Secure output:** Jinja2 escapes generated text before it is placed in the page.

5. Limitations

- The local model produces short answers and is less accurate on complex topics.
- Four of five features need internet access and Gemini quota.
- Markdown from Gemini is displayed as plain text, not rendered.
- There are no user accounts, so history and scores are not saved.
- The quiz is fixed at three questions and cannot be scored in the page.
- There is no loading indicator, and the first concept request is slow while the model loads.
- AI-generated content may contain mistakes and should be checked against reliable sources.

6. Future Enhancements

- Render Markdown in the response box and add a copy button.
- Add a loading indicator and stream answers as they are generated.
- Make quizzes interactive with scoring, adjustable length and difficulty.
- Add user accounts with history, bookmarks and progress tracking.
- Accept uploaded PDF or Word notes for summaries and quizzes.
- Add multilingual support and text-to-speech for accessibility.
- Cache repeated requests and add rate limiting to control cost.
- Offer GPU support or a larger local model for better offline explanations.

7. Final Remarks

EduGenie shows how language models can become a practical study companion with a compact, well-organised codebase. By combining a free local model with a stronger cloud model, it gives learners explanations, answers, summaries, quizzes and study roadmaps in one place. With personalisation, progress tracking and richer interaction added, it can grow from a working prototype into a complete AI-powered learning platform.